

Executive Summary

PROJECT OVERVIEW AND KEY FINDINGS

This project developed and evaluated a satellite-image-based pipeline for warehouse identification using sponsor-provided warehouse coordinates and publicly available geospatial data. The primary objective was to determine whether satellite imagery and deep learning could support scalable warehouse discovery and lead generation for logistics applications. Because the original dataset contained only positive warehouse locations, a major focus of the project was the construction of realistic and visually challenging non-warehouse negative samples using OpenStreetMap (OSM) data. Spatially separated evaluation later revealed that random train-test splitting substantially inflated apparent model performance due to geographic leakage.

The final dataset combined 4,205 cleaned warehouse locations with 1,528 negative samples collected from semantically meaningful categories such as industrial zones, commercial buildings, and wholesale facilities. Satellite images were retrieved using the Google Static Maps API and used to train binary image classifiers that distinguished warehouses from non-warehouse locations.

Multiple model architectures and dataset-construction strategies were evaluated, including YOLOv8 classifiers, ResNet-18 baselines, density-aware negative sampling, regularization schemes, and both random and spatial train-test splitting approaches. Across all experiments, YOLO-based classifiers consistently outperformed the ResNet baselines. The retained model, a YOLOv8s classifier trained on the `density_v2` dataset with stronger regularization settings, achieved the strongest balance between recall, precision, and ranking performance.

STANDARD PERFORMANCE METRICS AND VISUAL ANALYSIS

The retained model, **YOLO Run2 density_v2**, was chosen based on its performance in validation PR-AUC, recall, and macro-F1. Below is the final test-set performance achieved after temperature calibration:

Threshold Type	Recall	Precision	Macro F1	PR-AUC
Sensitive	0.9406	0.9577	0.9065	0.9888
Balanced	0.9406	0.9577	0.9065	0.9888
Aggressive	0.9857	0.8954	0.8654	0.9888

TABLE 1 Test-set performance metrics for the selected model.

The model also demonstrated strong operational ranking behavior. As shown in Figure 1, the top-ranked warehouse candidates achieved perfect precision for the top 10, 25, 50, 100, 200, and 300 ranked predictions on the held-out benchmark. This indicates that the system is highly effective at prioritizing likely warehouse locations for downstream business review workflows. Error analysis further revealed that the model is most challenged by “wholesale” commercial buildings, which share high visual similarity with warehouses. Prediction behavior across the selected operating thresholds is summarized in Figure 2.

SPATIAL GENERALIZATION GAP

Under random train-test splitting, the retained model achieved a macro F1-score of 0.9616. However, when geographically separated spatial splits were used to prevent nearby locations from appearing across training and testing partitions, macro F1-score decreased to 0.8965. This performance gap between the random split and the spatial split indicates that random splitting introduced geographic leakage and inflated apparent model performance. The spatially separated evaluation therefore provides a more realistic estimate of deployment performance in previously unseen geographic regions.

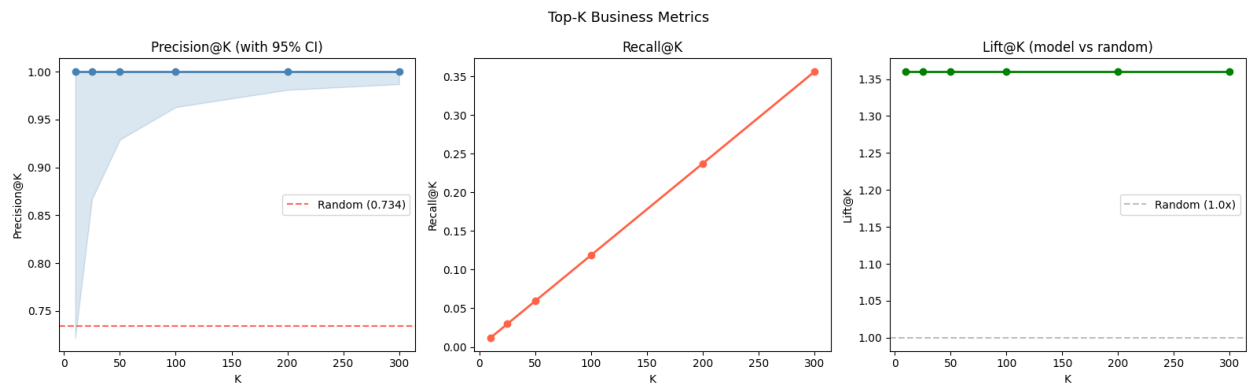


FIGURE 1 Top- K ranking performance for warehouse retrieval.

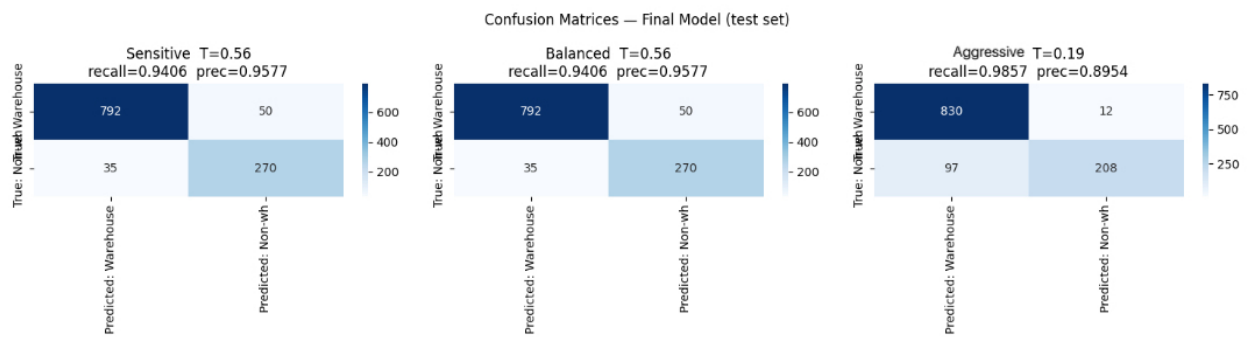


FIGURE 2 Confusion matrices across the selected operating thresholds.

From Coordinates to Sales Lead: Loadsmart Project Final Report

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STATEMENT OF SIGNIFICANCE (RELEVANCE OF PROJECT)

This report is of interest to Loadsmart because it investigates a practical pipeline for using publicly available geospatial information and satellite imagery to support warehouse-location intelligence. Accurate warehouse data can improve logistics planning, facility visibility, and broader supply-chain decision-making, but such data are difficult to maintain manually at scale. Our study is relevant to the sponsor because it evaluates how warehouse coordinates can be transformed into an image-based classification dataset and how different negative-sampling strategies affect downstream model behavior. Rather than presenting a final production-ready system, the report provides an internal benchmark, a structured comparison of data-construction choices, and evidence about which types of non-warehouse sites remain difficult to distinguish from warehouses. These findings are useful for understanding the feasibility, limitations, and next steps of satellite-imagery-based warehouse identification in an applied logistics setting.

AUTHOR CONTRIBUTION STATEMENT

The authors confirm contribution to the paper as follows: Study conception and design: Jasper Liu, Lavinia Xu, Iris Zhang, and Zi Zhu; Data preparation: Lavinia Xu and Jasper Liu; Analysis and interpretation of results: Jasper Liu, Lavinia Xu, Iris Zhang, and Zi Zhu; Draft manuscript preparation: Iris Zhang and Zi Zhu.

All authors reviewed the results and approved the final version of the manuscript.

INTRODUCTION

Efficient identification of warehouse facilities is an important capability for digital freight and logistics platforms such as Loadsmart. Reliable warehouse location information can support carrier matching, facility intelligence, route planning, and broader supply-chain decision-making. In practice, however, warehouse databases are difficult to maintain manually at scale because records may be incomplete, outdated, or inconsistent across regions. Recent advances in remote sensing and deep learning have made image-based recognition from satellite and aerial imagery increasingly feasible, but applied performance depends heavily on how the training data are constructed and evaluated rather than on model architecture alone (1, 2).

This project develops and evaluates a warehouse identification pipeline built from sponsor-provided warehouse coordinates and satellite imagery. The source data consist of latitude-longitude pairs corresponding to known warehouse locations. These coordinates are not used as direct predictive variables in a conventional tabular-learning setting; instead, they serve as geographic anchors for retrieving satellite tiles that can be passed to an image classifier. The resulting task is therefore framed as an image-based binary classification problem: given a satellite tile centered on a candidate location, predict whether the image contains a warehouse or a non-warehouse site. This framing is consistent with the broader remote-sensing scene-classification literature, where high-resolution aerial and satellite imagery are used to distinguish visually similar built-environment categories (2, 3).

The central technical challenge is that the sponsor data provide only positive warehouse examples. A usable binary classification dataset therefore requires construction of a realistic negative class. This is not a trivial preprocessing step. If negative examples are too easy, downstream model performance can be artificially inflated without reflecting operational difficulty. If they are too noisy or semantically inconsistent, the model may learn unstable decision boundaries. More generally, recent work on data-centric machine learning has emphasized that data design, curation, and validation are often as consequential as model selection, especially in applied settings where labels are incomplete or assembled from heterogeneous sources (4). For this reason, our study is structured around a data-construction question as much as a model-training question: how should non-warehouse negatives be sampled so that the resulting image classification problem is realistic, informative, and useful for downstream warehouse detection?

To address this question, we built a complete pipeline that combines geospatial processing, negative-sample construction, satellite image retrieval, supervised image classification, probability calibration, and business-oriented lead generation. We first cleaned and deduplicated the warehouse coordinate data, reducing the initial coordinate table to 4,205 unique positive warehouse points. We then characterized local spatial structure among positives using DBSCAN and assigned warehouse points to density types representing isolated locations, small clusters, and dense clusters. This density information motivated a comparison between two negative-sampling strategies. The first was a baseline global strategy that applied one common set of search and exclusion rules across all positive seeds. The second was a density-aware strategy that varied search distance and exclusion parameters according to the local clustering pattern of the warehouse seeds. Negative coordinates were constructed using OpenStreetMap categories and spatial exclusion rules, then paired with Google satellite imagery to create the final image dataset.

After dataset construction, we generated both random and spatial train-validation-test splits and trained multiple image classifiers. The principal models evaluated in the benchmark were YOLOv8 image classification models and a ResNet-18 baseline. We compared negative-sampling

strategy, model family, regularization strength, label smoothing, offline augmentation, and split design through the main training pipeline and a separate ablation study. In addition to overall classification metrics, we examined category-level error patterns, reviewed representative predictions with image-level explanation tools, calibrated prediction probabilities, and translated model scores into operational outputs such as top- K ranking metrics, city-level opportunity summaries, and warehouse lead CSV files.

The final results show several important patterns. First, the pipeline successfully converts point-based warehouse records into a satellite-image classification benchmark with interpretable metadata, train-validation-test splits, and category-level negative labels. Second, YOLO-based classifiers consistently outperform the ResNet-18 baseline in the evaluated setting. Third, spatially aware evaluation revealed that random train-test splitting can substantially inflate performance due to geographic leakage between nearby locations. Finally, the ablation study shows that additional pipeline complexity does not uniformly improve downstream performance: density-aware negative sampling, stronger regularization, label smoothing, and offline augmentation did not consistently improve validation metrics across all settings. These findings suggest that design choices in data construction and evaluation may be as important as model architecture itself in warehouse-identification tasks built from geospatial anchors.

The project matters in both practical and methodological terms. Practically, it provides Loadsmart with a framework for expanding or validating warehouse intelligence using publicly available geospatial information and satellite imagery. Methodologically, it contributes a data-centric case study showing that the quality and realism of negative samples are central to remote-sensing classification tasks built from positive anchors only. Rather than claiming a final production-ready warehouse detector, this report presents an internal benchmark, a structured comparison of negative-sampling choices, an ablation-based evaluation of modeling decisions, and a prototype lead-ranking workflow designed to support scalable warehouse discovery.

The remainder of this report is organized as follows. The methodology section describes data cleaning, spatial density labeling, negative-sampling design, image retrieval, model training, evaluation, calibration, and lead-generation outputs. The findings section presents dataset statistics, comparative model performance, ablation results, category-level error analysis, calibration behavior, and business-oriented ranking metrics. The discussion section then considers the practical implications of the results, the main limitations of the current benchmark, and directions for improving robustness and external validity.

METHODOLOGY

Data Source and Positive Sample Preparation

The sponsor-provided data contained known warehouse locations represented by latitude-longitude pairs. These coordinates were not used as tabular predictors; instead, they served as anchors for satellite image retrieval. The task was framed as binary image classification: given a satellite tile centered on a candidate location, classify the image as `warehouse` or `non_warehouse`. This framing follows remote-sensing scene-classification practice, where coordinates define image samples rather than the final feature space (2, 5).

The original sponsor table contained 4,702 warehouse coordinate entries, including 942 duplicate rows. After exact latitude-longitude de-duplication, the final positive set contained 4,205 unique warehouse locations. This cleaned table, `positive_points_cleaned_dedup.csv`, was used as the definitive positive source for sampling, image retrieval, splitting, and model training.

To characterize warehouse spatial context, we applied DBSCAN to the cleaned positive coordinates after projecting them from WGS84 into metric space. A parameter sweep considered neighborhood radii of 7.5 km, 10 km, and 15 km crossed with minimum sample thresholds of 5, 8, and 12. We selected `eps = 10 km` and `min_samples = 5` because the resulting clustering structure balanced isolated sites and meaningful geographic clusters. DBSCAN noise points were labeled `isolated`; clustered points in clusters of at least 20 warehouses were labeled `dense_cluster`; remaining clustered points were labeled `small_cluster`. The resulting table, `positive_points_density_labels.csv`, was later used for density-aware negative sampling and spatial splitting.

Negative Sample Construction

Because the sponsor data contained only warehouse locations, the negative class had to be constructed externally. We used OpenStreetMap (OSM) as a source of labeled points of interest and land-use information (6). The objective was to create non-warehouse examples that were geographically plausible, semantically interpretable, and visually challenging.

The soft-negative pipeline queried OSM candidates around sampled warehouse seeds using `features_from_point`. OSM geometries were standardized to point representations, converted to WGS84 coordinates, and filtered using a positive-site exclusion rule. All positive warehouse points were buffered in projected metric space, and any OSM candidate inside the exclusion union was removed. Remaining candidates were de-duplicated by rounded latitude-longitude coordinates and sampled down to target counts.

The main soft-negative set contained ten categories: fuel, parking, restaurant, hospital, hotel, supermarket, school, park, playground, and sports center. Each category targeted 100 retained locations. Two strategies were retained. The control strategy, `baseline_global`, used the same 2.5 km search radius and 1.0 km exclusion buffer for all warehouse density contexts. The density-aware strategy, `density_v2`, adjusted parameters by seed density type: `isolated` seeds used a 5.0 km search radius and 0.8 km exclusion buffer, `small_cluster` seeds used 3.0 km and 1.2 km, and `dense_cluster` seeds used 1.8 km and 1.6 km. For each category, seed allocation was fixed at 20 `isolated`, 15 `small-cluster`, and 15 `dense-cluster` warehouse seeds, giving 50 positive seed locations per category.

Initial soft-negative sampling produced 911 negatives for `density_v2` model and 900 for `baseline_global` model, leaving some categories below the 100-location target. A top-up stage reused the same exclusion and de-duplication logic but expanded selected OSM tag definitions, such as adding `clinic` and `doctors` for hospital, `motel` and `guest_house` for hotel, `wholesale` for supermarket, and `stadium` and `fitness_centre` for sports center. Top-up queries used 40 positive seeds and strategy-specific flat search parameters. After this step, both strategies reached 100 observations in each soft-negative category.

A hard-negative stage was added to increase visual difficulty. Five warehouse-like categories were queried from OSM: industrial, factory, large retail, commercial, and wholesale. These were sampled with 40 positive seeds, a 4.0 km search radius, and a 1.0 km exclusion buffer, then appended to both strategy-specific CSVs after coordinate de-duplication. Because wholesale and factory remained under-covered, a final extended top-up used expanded tags, a 5.0 km search radius, a 1.0 km exclusion buffer, and 60 positive seeds. The final retained counts were 148 wholesale and 80 factory examples. Both final negative metadata files, `neg_density_v2_filled.csv` and `neg_baseline_global_filled.csv`, contained 1,528 negative locations. This construction

process was designed to balance semantic realism, geographic relevance, and visual difficulty. Because OSM candidates came from live queries, the reported experiments are tied to the archived metadata tables used for image retrieval rather than to regenerated OSM query results.

Satellite Image Retrieval and Dataset Assembly

Each finalized coordinate was converted into a satellite tile using the Google Static Maps API with zoom level 18, image size 640×640 , and satellite map type. Dataset assembly was performed separately for `density_v2` and `baseline_global`. For each strategy, 4,205 positive warehouse rows were combined with 1,528 negative rows, producing 5,733 observations. Positive rows were labeled `warehouse`, negative rows were labeled `non_warehouse`, and each row received an image filename and metadata record.

Each master table was partitioned in two ways. First, a fixed-seed stratified random split used a 60/20/20 ratio. This split supported the main YOLO runs, label-smoothing runs, and ResNet baselines. It produced 3,439 training observations, 1,147 validation observations, and 1,147 test observations. The training split contained 2,522 positives and 917 negatives; validation contained 841 positives and 306 negatives; test contained 842 positives and 305 negatives.

Second, a spatial split evaluated whether random splitting inflated performance through geographic proximity. Positive warehouses were grouped by DBSCAN cluster labels, isolated positives received unique groups, and negatives received unique groups. This group-based split prevented clustered warehouse positives from crossing train and evaluation partitions. The spatial split produced 3,561 training observations, 1,086 validation observations, and 1,086 test observations. Training contained 2,629 positives and 932 negatives, while both validation and test contained 788 positives and 298 negatives.

Images were written to directory structures compatible with YOLO and PyTorch loaders, including `train/warehouse`, `train/non_warehouse`, `val/warehouse`, `val/non_warehouse`, `test/warehouse`, and `test/non_warehouse`. Manifest files preserved coordinates, labels, split assignments, filenames, negative categories, and download status. Download logs recorded API responses, and all retrieved images were validated with the Python Imaging Library. No corrupt images remained after validation in the recorded benchmark.

Because the training data contained more warehouse than non-warehouse examples, augmentation was applied only to negative training images. Three augmentation combinations based on flips, 90-degree rotation, brightness adjustment, and contrast adjustment were added to the original negative training tiles. Validation and test data were not augmented. This produced 3,668 non-warehouse and 2,522 warehouse training images in the random split, and 3,728 non-warehouse and 2,629 warehouse training images in the spatial split.

Model Training

Training compared negative-sampling strategy, model family, regularization, and split design. The primary model family was YOLOv8 image classification, implemented through Ultralytics and initialized from pretrained `yolov8s-cls.pt` weights. For each negative-sampling strategy, we trained three random-split YOLO configurations: Run 1 with dropout 0.3 and weight decay 10^{-4} ; Run 2 with dropout 0.5 and weight decay 10^{-3} ; and Run 2 + Label Smoothing, which used the Run 2 regularization settings with label smoothing of 0.1. All random-split YOLO runs used 50 epochs, image size 640, batch size 32, early stopping patience of 10 epochs, and a fixed random seed.

One spatial-split YOLO model was also trained for each strategy using the Run 2 regularization settings. These models evaluated whether random-split performance was affected by geographic similarity between training and evaluation samples. In total, the YOLO benchmark contained eight models: Run 1, Run 2, Run 2 + Label Smoothing, and Spatial Split for both `density_v2` and `baseline_global`. The YOLO trainer automatically selected AdamW in the recorded runs.

We also trained ResNet-18 baselines for both strategies. Each ResNet used ImageNet-pretrained weights with a two-class output head. Images were resized to 224×224 and normalized with ImageNet statistics. Training used Adam, learning rate 10^{-3} , weight decay 10^{-4} , cross-entropy loss, ReduceLRonPlateau scheduling, up to 50 epochs, and early stopping patience of 10 epochs.

Evaluation, Lead Ranking, and Interpretability

Evaluation combined the model-selection pipeline with the downstream testing and lead-generation pipeline. The evaluation pipeline compared trained models, selected the retained classifier, calibrated scores, tuned thresholds, and performed test-set error analysis. The downstream evaluation pipeline restored the retained model and translated it into deployment-oriented outputs such as top- K metrics, city summaries, lead tiers, and candidate CSV files.

Model selection was performed on the validation split. Selection emphasized PR-AUC, warehouse recall, warehouse precision, non-warehouse F1, and macro-F1 rather than accuracy alone. Each model was temperature-calibrated on validation predictions, with expected calibration error recorded before and after calibration. The retained model was chosen from non-spatial candidates using validation PR-AUC as the primary criterion, recall and macro-F1 as secondary criteria, and a minimum precision constraint to avoid high-recall but low-precision solutions.

After model selection, a hard-negative fine-tuning check was performed for 10 additional epochs on warehouse-like negative categories. Fine-tuned weights were adopted only if validation macro-F1 improved. For the final model, three operating thresholds were tuned on calibrated validation probabilities: `sensitive` was chosen as the highest-precision threshold among thresholds with warehouse recall ≥ 0.95 , `balanced` was selected by maximizing warehouse F1, and `aggressive` was chosen as the lowest threshold satisfying warehouse precision ≥ 0.90 . These thresholds were saved to `threshold_config.csv` and applied to the held-out test set.

The final test report included warehouse recall, warehouse precision, warehouse F1, non-warehouse F1, macro-F1, and test PR-AUC. Bootstrap resampling with 1,000 replicates was used to estimate variability for recall, precision, and macro-F1. Confusion matrices were generated for the `sensitive`, `balanced`, and `aggressive` thresholds. Category-level error analysis used saved `neg_category` labels to compute false-positive rates and Wilson confidence intervals for each negative category. False negatives were summarized separately as borderline or confident misses. Scores were also converted into three operational tiers: Core Warehouse Candidate, Secondary Logistics Candidate, and Deprioritized.

For interpretability, representative true positives, true negatives, false positives, and false negatives were selected for visual review. Grad-CAM-style maps were generated using the retained YOLO classifier’s feature-visualization mechanism, and image-based SHAP explanations were computed on the same representative subset using blurred-mask perturbations and 224×224 inputs.

The lead-ranking pipeline extended evaluation into deployment-oriented candidate prioritization. It restored the final weights, restored the `density_v2` dataset, loaded validation-derived

thresholds, refit temperature scaling on validation data for self-containment, and generated calibrated test scores. Candidates were ranked to compute Precision@ K , Recall@ K , and Lift@ K for $K = 10, 25, 50, 100, 200, 300$, and the full test set. Precision@ K included Wilson confidence intervals, while Lift@ K compared model ranking against random selection.

The lead-generation pipeline also assigned test coordinates to nearby U.S. cities using Natural Earth populated-place data and aggregated scores into city-level opportunity summaries. It further demonstrated scoring new candidate buildings from Overture Maps in a pilot region, downloading matching satellite tiles and applying the final model. Known and newly scored candidates were combined into `warehouse_leads.csv`, with coordinates, scores, tiers, source labels, Google Maps links, model version, and scoring date. A second file, `warehouse_leads_adjusted.csv`, applied a placeholder prior adjustment for the lower expected warehouse prevalence in real-world search pools.

Code and Reproducibility

All experiments, dataset-construction pipelines, training procedures, evaluation scripts, and lead-generation workflows were organized into modular Jupyter notebooks and archived outputs within the accompanying project repository. The repository separates negative-sampling development, dataset construction, model training, evaluation, and ablation analysis into reproducible stages with corresponding saved artifacts and metadata outputs.

FINDINGS

Model Comparison

Table 1 summarizes the validation performance of all trained models. The benchmark included three YOLOv8s-cls variants (Run 1 with weaker regularization, Run 2 with stronger regularization, and Run 2 with stronger regularization and label smoothing), together with ResNet-18 baselines trained on both the `density_v2` and `baseline_global` datasets. Performance was evaluated using PR-AUC, warehouse recall, warehouse precision, non-warehouse F1, macro F1, and calibration temperature.

Model	PR-AUC	Recall	Precision	Non-wh F1	Macro F1	T
YOLO Spatial <code>density_v2</code>	0.9885	0.9683	0.9397	0.8706	0.9122	2.516
YOLO Run2 <code>density_v2</code>	0.9883	0.9584	0.9494	0.8709	0.9124	2.097
YOLO Run2+LS <code>density_v2</code>	0.9883	0.9584	0.9494	0.8709	0.9124	2.097
YOLO Run1 <code>density_v2</code>	0.9863	0.9429	0.9486	0.8525	0.8991	1.824
YOLO Spatial <code>baseline_global</code>	0.9841	0.9569	0.9390	0.8571	0.9025	3.074
YOLO Run1 <code>baseline_global</code>	0.9838	0.9608	0.9384	0.8547	0.9021	3.849
YOLO Run2 <code>baseline_global</code>	0.9801	0.9370	0.9392	0.8306	0.8844	3.140
YOLO Run2+LS <code>baseline_global</code>	0.9801	0.9370	0.9392	0.8306	0.8844	3.140
ResNet <code>baseline_global</code>	0.9736	0.9572	0.8703	0.7045	0.8081	1.064
ResNet <code>density_v2</code>	0.9730	0.8823	0.9298	0.7634	0.8344	0.920

Table 1 Validation performance comparison across models.

Initial model selection on the validation set prioritized PR-AUC and warehouse recall while enforcing a minimum precision constraint. Among the non-spatial benchmark models, YOLO Run2 `density_v2` was selected as the retained model. It achieved a PR-AUC of 0.9883, ware-

house recall of 0.9584, and macro F1 of 0.9124. Across all evaluated settings, YOLO-based classifiers consistently outperformed the ResNet-18 baselines, indicating that the YOLOv8 classification architecture provided stronger feature extraction and decision boundaries for this satellite-imagery task.

A key finding emerged from the spatial leakage analysis. When the retained YOLO Run2 density_v2 configuration was evaluated under spatial splitting, macro F1 decreased from 0.9616 on the random test split to 0.8965 on the spatially separated test split. This gap indicates that random train-test partitioning allowed geographically proximate samples to appear across splits, producing inflated performance estimates. The spatially separated evaluation therefore provides a more realistic estimate of generalization to unseen geographic regions.

Ablation Study

To further evaluate the design choices behind the final pipeline, we conducted an ablation study across model family, input resolution, augmentation, regularization, label smoothing, and negative-sampling strategy.

The ablation results supported the primary architectural conclusions from the benchmark experiments. YOLO-based classifiers consistently outperformed the ResNet-18 baseline, and the YOLOv8s configuration at 320-pixel resolution outperformed the smaller YOLOv8n model at 640-pixel resolution. These findings suggest that additional model capacity was beneficial for the warehouse-identification task despite the lower image resolution.

The ablation study also clarified which forms of additional pipeline complexity were most beneficial in the current benchmark. The strongest logged validation performance was achieved by the YOLOv8s classifier trained using the density_v2 negative set with stronger regularization. However, density-aware sampling, label smoothing, stronger regularization, and offline augmentation did not uniformly improve validation metrics across all configurations. Rather than weakening the benchmark, these findings help narrow the recommended design space: a relatively simple YOLOv8s pipeline already achieved strong internal performance, while additional sampling and training modifications should be treated as tunable extensions rather than guaranteed improvements.

Probability Calibration and Threshold Tuning

Temperature scaling was applied to improve probability calibration for the retained YOLO Run2 density_v2 classifier. The optimal validation-fitted temperature parameter was $T = 2.0967$. Calibration substantially reduced expected calibration error (ECE). Validation ECE improved from 0.0455 to 0.0199, while test ECE improved from 0.0452 to 0.0154. As shown in Figure 1, temperature scaling noticeably improved alignment between predicted confidence and empirical accuracy across probability bins. These results indicate that the calibrated probabilities more accurately reflected empirical prediction uncertainty.

Three operational thresholds were selected using calibrated validation probabilities to support different downstream objectives. Table 2 summarizes the selected operating points.

The sensitive threshold was designed to maintain recall above 0.95 while maximizing precision under that constraint. The balanced threshold was selected by maximizing warehouse F1-score. In the recorded benchmark, these two operating criteria converged to the same threshold value ($T = 0.556$), indicating that the F1-optimal operating point also satisfied the desired high-recall constraint. The aggressive threshold lowered the decision boundary to prioritize broader

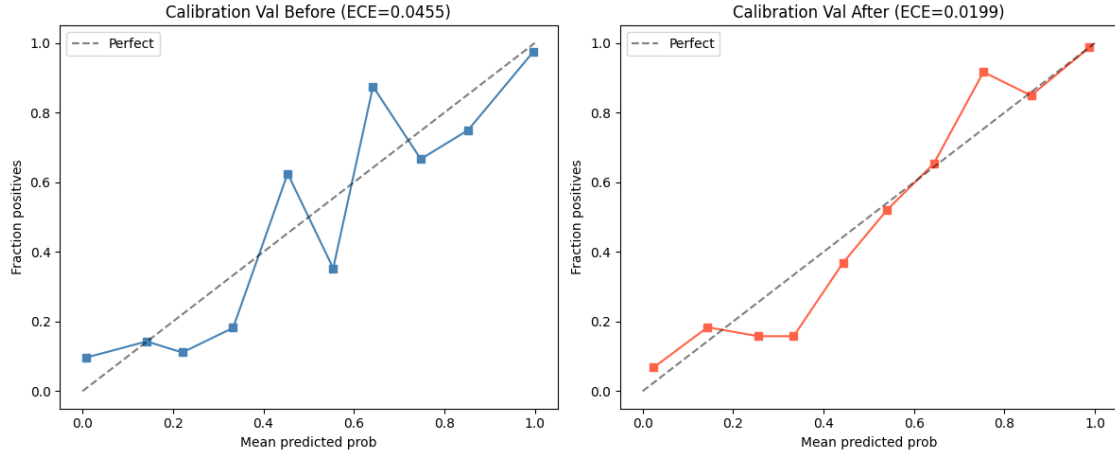


Figure 1 Reliability diagram before and after temperature scaling calibration.

Threshold Type	Threshold	Recall	Precision	F1-score
sensitive	0.556	0.9512	0.9627	0.9569
balanced	0.556	0.9512	0.9627	0.9569
aggressive	0.191	0.9738	0.9000	0.9355

Table 2 Threshold tuning results on the validation set.

warehouse retrieval coverage. This operating point increased warehouse recall to 0.9738 at the cost of lower precision (0.9000). As illustrated in Figure 2, lowering the classification threshold increased recall while reducing precision, reflecting the expected tradeoff between retrieval coverage and false-positive control. Operationally, the aggressive threshold is appropriate for exploratory lead generation and high-recall search scenarios where additional false positives are acceptable.

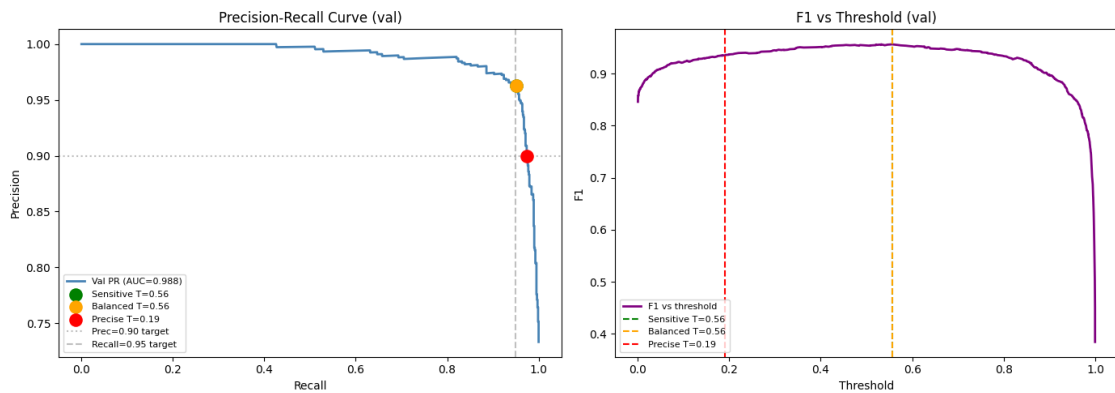


Figure 2 Performance tradeoffs across classification thresholds.

Final Test Performance

The retained YOLO Run2 density_v2 classifier, calibrated using the validation-fitted temperature parameter, was evaluated on the held-out test set using the selected operating thresholds. Table 3 summarizes the final test performance.

Threshold	Recall	Precision	Macro F1	PR-AUC
sensitive	0.9406	0.9577	0.9065	0.9888
balanced	0.9406	0.9577	0.9065	0.9888
aggressive	0.9857	0.8954	0.8654	0.9888

Table 3 Final test-set performance of the retained model.

The retained model maintained strong ranking performance on the held-out test set, achieving a PR-AUC of 0.9888 across all operating thresholds. At the balanced threshold, the classifier achieved 0.9406 recall and 0.9577 precision, indicating strong discrimination between warehouse and non-warehouse imagery. The aggressive threshold further increased warehouse recall to 0.9857 while reducing precision, reflecting the expected tradeoff between retrieval coverage and false-positive control. The resulting prediction behavior at the balanced threshold is summarized in the confusion matrix shown in Figure 3.

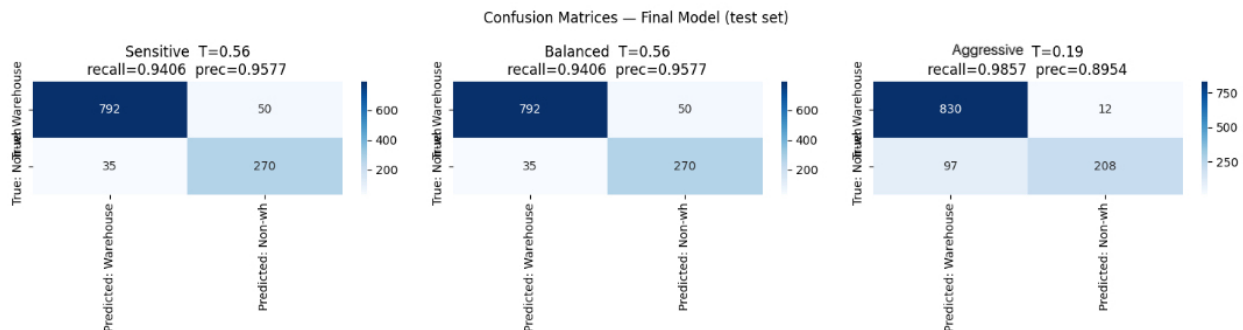


Figure 3 Confusion matrices across the selected operating thresholds.

Error Analysis

False-positive rates were analyzed across OpenStreetMap categories using the retained model. As shown in Figure 4, the highest false-positive rate was observed for wholesale regions, where 58.3% of samples were incorrectly classified as warehouses. This result suggests that the model learned warehouse-like logistics morphology rather than memorizing narrow category-specific features. Large industrial facilities, wholesale buildings, and commercial distribution sites often share visual characteristics with warehouses, including large rectangular roof structures, loading areas, and adjacent pavement.

At the balanced threshold, 50 warehouse samples were missed, corresponding to 5.9% of warehouse instances in the test set. Approximately 22% of false negatives were borderline misses located within 0.10 of the decision threshold, while 66% received substantially lower confidence scores. The borderline misses are recoverable under the aggressive operating point, whereas the

lower-confidence misses likely correspond to visually atypical warehouses with uncommon roof colors, obstructed footprints, or unusual surrounding context.

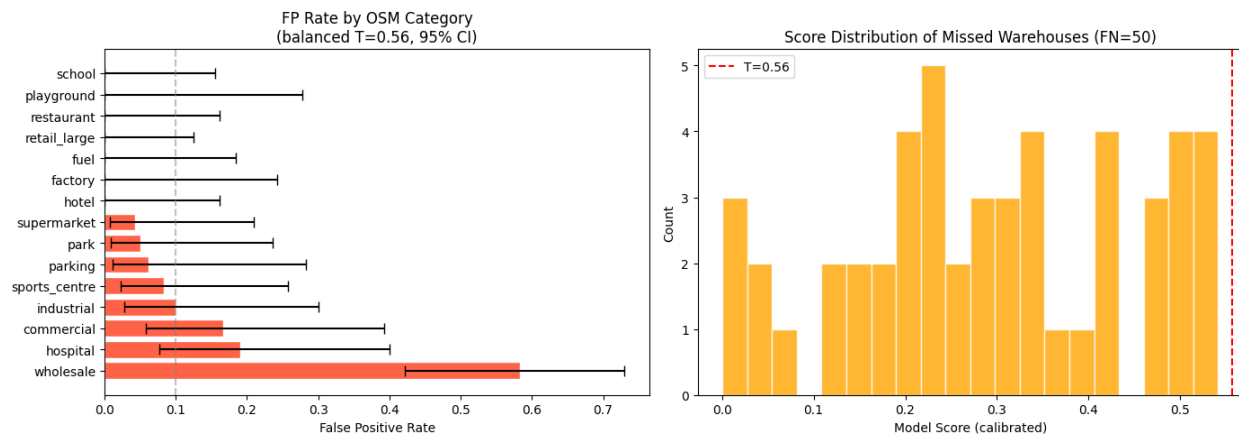


Figure 4 Category-level false-positive rates and score distribution of missed warehouses.

Tiered Classification Strategy

To support downstream prioritization workflows, prediction scores were divided into three operational tiers summarized in Table 4. Core Warehouse Candidate represents highly confident warehouse detections suitable for immediate review, while Secondary Logistics Candidate captures more ambiguous logistics-related facilities that may benefit from additional downstream screening or human validation.

Tier	Samples	Precision	95% CI
Core Warehouse Candidate	827	0.958	[0.942, 0.969]
Secondary Logistics Candidate	100	0.380	[0.291, 0.478]
Deprioritized	220	0.055	[0.031, 0.093]

Table 4 Tiered classification performance.

The Core Warehouse Candidate tier achieved a precision of 95.8%, demonstrating that the retained model can effectively prioritize highly probable warehouse candidates for downstream workflows. The Secondary Logistics Candidate tier captured more ambiguous industrial and logistics-related structures, while the Deprioritized tier contained predominantly non-warehouse samples. Importantly, the Deprioritized tier still contained a small number of true warehouse examples. Consequently, deprioritization should not be interpreted as definitive exclusion in scenarios where maximum warehouse recall is operationally important.

Model Interpretability

We further examined model interpretability using SHAP visualizations generated from representative true-positive, true-negative, false-positive, and false-negative predictions produced by the retained YOLO Run2 density_v2 classifier.

Correctly classified warehouse examples consistently highlighted large rectangular roof structures, loading regions, and surrounding pavement as strong positive contributors. Conversely,

vegetation, curved roads, parking layouts, and residential context frequently contributed negative evidence toward warehouse predictions. False-positive examples demonstrated that the model was strongly activated by visually warehouse-like commercial structures, particularly large flat roof segments and unobstructed industrial pavement. False-negative examples suggested a different failure mode: although some warehouse-like structures were identified correctly, surrounding contextual features reduced the final prediction score below the classification threshold.

Since these explanations were generated from a limited qualitative subset, they should be interpreted as illustrative examples rather than exhaustive feature-attribution analysis.



Figure 5 Representative true-positive SHAP visualization.



Figure 6 Representative false-positive SHAP visualization.

Operational Evaluation and Candidate Discovery

Precision@K and Lift Analysis

To evaluate operational usefulness for sales and logistics workflows, ranked retrieval metrics were computed using calibrated warehouse probabilities. Candidate locations were sorted according to predicted warehouse probability, and Precision@K, Recall@K, and Lift@K were calculated. The baseline warehouse prevalence in the held-out test set was 0.7341.

Results showed extremely strong precision among the highest-ranked candidates. As shown in Figure 7, the top 10, 25, 50, 100, 200, and 300 ranked predictions all achieved perfect precision on the held-out test set. Relative to the baseline warehouse prevalence, this corresponds to an approximate Lift@K of $1.36\times$ across the highest-ranked retrieval ranges. These findings indicate that the ranking pipeline effectively prioritizes highly probable warehouse candidates, allowing downstream reviewers to focus attention on the most promising facilities first.

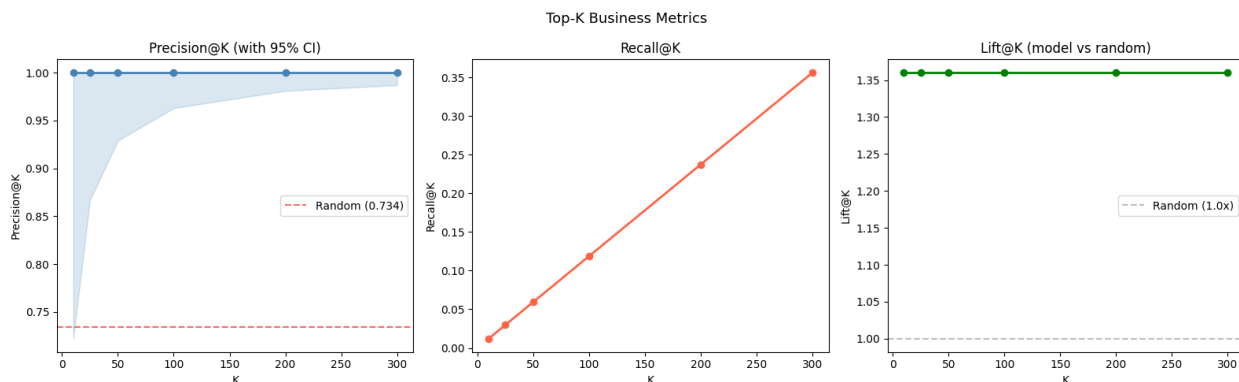


Figure 7 Top-K ranking performance for warehouse retrieval.

Regional Opportunity Analysis

A city-level regional analysis was conducted to evaluate whether predicted warehouse probabilities aligned with known geographic warehouse-density patterns. A total of 420 U.S. cities with populations greater than 50,000 were included. Warehouse candidate locations were assigned to nearby metropolitan areas using nearest-neighbor geographic matching. A statistically significant positive relationship was observed between mean model score and regional warehouse density, with Spearman correlation coefficient $\rho = 0.647$ and $p < 0.001$. This result suggests that the model captured meaningful regional logistics structure rather than random spatial variation.

Based on the proposed opportunity score, which is defined as mean score multiplied by core-candidate count, the highest-ranked metropolitan areas included Columbus, Aurora, Portland, Glendale, and Houston (Figure 8).

Overture Maps Candidate Discovery

To evaluate generalization beyond the benchmark dataset, we piloted a candidate-discovery workflow using Overture Maps commercial-building polygons in the Dallas–Fort Worth metropolitan area. A total of 100 previously unseen commercial-building candidates with estimated footprint area greater than 2,000 square meters were processed through the retained model pipeline. The resulting score distribution had mean score 0.5037, median score 0.5191, and maximum score 0.9995,

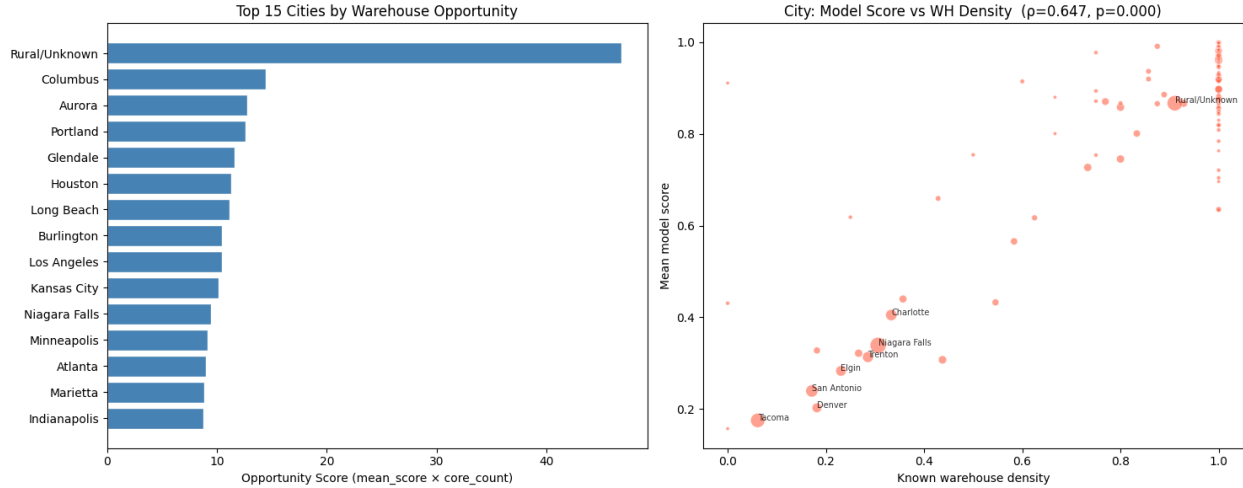


Figure 8 Regional opportunity scores across metropolitan areas.

indicating that the pipeline successfully assigned a broad range of calibrated warehouse probabilities to novel commercial structures outside the original benchmark set.

Final Leads and Real-World Prior Adjustment

The final pipeline compiled a unified lead table containing 1,247 scored candidates. Based directly on calibrated model probabilities, 998 candidates (80.0%) satisfied the Core Warehouse Candidate threshold. Among the 100 newly discovered Overture candidates, 72 were classified as core warehouse candidates.

Tier	Count	Percentage
Core Warehouse Candidate	998	80.0%
Secondary Logistics Candidate	0	0.0%
Deprioritized	249	20.0%

Table 5 Warehouse lead tier distribution.

Since the benchmark dataset contained approximately 73% warehouse examples, the calibrated probabilities reflected a substantially higher warehouse prevalence than would be expected in real-world commercial-building search pools. To address this mismatch, posterior probabilities were adjusted using an assumed real-world warehouse prevalence of 8%.

After prior adjustment, 58.7% of candidates (732 locations) remained classified as Core Warehouse Candidate. This adjustment substantially reduced optimistic prevalence assumptions while still preserving a large pool of high-confidence warehouse leads for downstream review.

DISCUSSION

Our results demonstrate that satellite-image classification is a feasible approach for transforming point-based warehouse records into a scalable warehouse-identification and candidate-ranking pipeline. Across the evaluated benchmark settings, YOLO-based classifiers consistently outperformed the ResNet-18 baselines, indicating that the YOLOv8 classification architecture pro-

vided stronger feature extraction and discrimination for high-resolution satellite imagery. The retained YOLO Run2 density_v2 configuration achieved strong validation and test performance while maintaining high warehouse recall and precision after calibration.

At the same time, the spatial-split experiments substantially changed how the benchmark results should be interpreted. Although random train-validation-test splits produced very strong internal metrics, evaluation under geographically separated splits revealed a meaningful performance decrease. This finding suggests that random partitioning allowed geographically proximate samples with similar visual context to appear across training and evaluation sets, producing optimistic estimates of generalization performance. Consequently, the spatial-split results likely provide a more realistic estimate of deployment behavior in previously unseen regions. More broadly, this result highlights that evaluation design is as important as model architecture in remote-sensing classification tasks built from geospatial anchors.

The ablation study further clarified which forms of pipeline complexity were most useful in the current benchmark. The experiments supported the broader architectural decision to use YOLO-based classification rather than a simpler ResNet baseline, and they showed that the larger YOLOv8s configuration provided stronger performance than the smaller YOLOv8n setting. However, several additional modifications, including density-aware sampling adjustments, stronger regularization, label smoothing, and offline augmentation, did not consistently improve validation metrics across all configurations. These findings suggest that additional pipeline complexity should not automatically be assumed to improve downstream performance. Instead, modifications should be evaluated empirically against realistic validation objectives.

The results also reinforce the importance of negative-set construction. Although the density-aware strategy was conceptually motivated, its improvements over the simpler baseline_global strategy were limited and inconsistent across evaluation settings. By contrast, the error analysis showed that visually warehouse-like categories such as wholesale, industrial, commercial, and factory produced substantially higher false-positive rates than unrelated land-use categories. This pattern suggests that semantic and visual similarity may be more important than geographic tailoring alone when constructing informative hard negatives. Future dataset improvements should therefore focus on expanding carefully reviewed warehouse-adjacent categories rather than increasing sampling complexity without clear empirical benefit.

From an operational perspective, false positives should not necessarily be interpreted as uniformly undesirable outputs. Many high-scoring false positives corresponded to logistics-adjacent facilities such as industrial buildings, wholesale properties, distribution sites, and large commercial structures. Although these facilities were not labeled as warehouses in the benchmark dataset, they may still represent commercially relevant properties for freight, logistics, or warehousing workflows. Consequently, some apparent classification errors may still correspond to useful downstream business leads. This distinction is important because the practical objective of the system is not solely binary classification accuracy, but scalable prioritization of warehouse-like facilities for downstream review.

The lead-ranking experiments further support this interpretation. Precision@K and Lift@K analysis showed that the retained model effectively prioritized likely warehouse candidates at the top of the ranked list. In practical deployment settings, this ranking behavior may be more operationally meaningful than overall accuracy because analysts or sales teams typically review only a limited subset of the highest-confidence candidates. The tiered candidate outputs, city-level opportunity summaries, calibrated scoring pipeline, and lead-generation tables were designed to support

this prioritization workflow rather than fully automated warehouse labeling.

Several limitations should nevertheless be acknowledged. First, the benchmark remains an internal evaluation rather than evidence of production readiness. The datasets were constructed from known warehouse coordinates and OSM-derived negative samples, and real-world candidate pools are likely to exhibit substantially lower warehouse prevalence and greater visual diversity. Second, although spatially separated evaluation was included, additional geographic holdout strategies across states, metropolitan areas, or entirely unseen regions would provide a stronger estimate of deployment robustness. Third, OSM category labels are imperfect proxies for true non-warehouse status, particularly for logistics-adjacent commercial properties. Finally, calibrated probabilities and threshold operating points remain sensitive to prevalence assumptions, requiring careful prior adjustment before operational deployment.

More broadly, the project should be interpreted as a reproducible internal benchmark and decision-support prototype rather than a final deployment-ready warehouse detector. The results demonstrate that a pipeline combining sponsor-provided coordinates, externally constructed negative samples, satellite imagery, calibration, and operational ranking can produce meaningful discrimination under controlled conditions. At the same time, the experiments highlight that data construction, evaluation design, calibration, and downstream business interpretation must all be considered together. Future work should therefore focus on stronger geographic holdout validation, expanded hard-negative datasets, improved prevalence-aware calibration, and systematic review of high-scoring false positives to determine whether they represent genuine errors, logistics-adjacent facilities, or commercially useful leads.

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